

Official Review Comments and Required Corrections

Candidate Network Crosswalk and Axis Minimum-Evidence Template

Purpose of this review

The documents shown in the screenshots appear to contain two related development datasets:

A Candidate Network Crosswalk intended to connect the 180 Living Biological Networks to the official Thirty-Nine Biological Axes and Twelve Domains.

An Axis Minimum-Evidence Template intended to define the minimum evidence required before each of the Thirty-Nine Biological Axes may be assessed or scored.

The work is moving in the correct direction. Converting the Network table from a visually formatted PDF into structured, machine-readable data is necessary and appropriate.

However, these files must currently be treated as:

status = CANDIDATE

authoritative = false

production_allowed = false

They are suitable for expert review, but they are not yet approved TSPI Master Registries.

This review is based on the visible screenshots. A final row-by-row approval of all 180 Networks requires the complete network_crosswalk_FULL_DRAFT.csv.

1. Overall assessment

What has been done correctly

The development team has correctly:

converted the 180-Network list into a structured table;

retained the original labels from the source document;

added proposed Axis and Domain mappings;

added mapping-confidence values;

included a biological scope or explanatory field;

created confirmation columns for expert review;

used the current A1–A39 Axis list in the Minimum-Evidence sheet;

included A35–A39 and removed CELA as an Axis;

kept proposed relationships separate from confirmed relationships.

These are appropriate engineering choices.

What remains incomplete

The current files still require:

semantic review of the proposed Network-to-Axis mappings;

clear separation between Primary and Secondary Axes;

automatic Domain derivation from the approved Primary Axis;

correction of several biologically inconsistent candidate mappings;

formal definition of the confidence values;

completion of the Network Edge Registry;

completion of the Minimum-Evidence rules for all 39 Axes;

version control and expert approval;

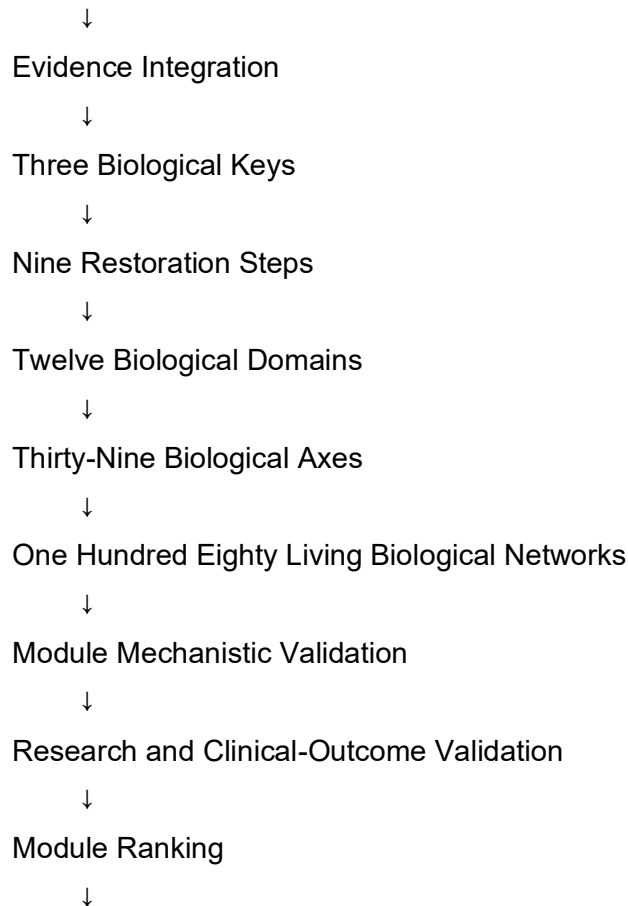
production gates preventing provisional data from affecting clinical recommendations.

2. Constitutional rule: the Thirty-Nine Axes remain central

The development team must not allow the Network Crosswalk to change the central architecture of TSPI.

The correct hierarchy remains:

Clinical Evidence



Physician Approval

The Thirty-Nine Axes are the primary biological assessment and Module-candidate layer.

The Networks are used to explain:

why an Axis is disturbed;

how biological failure propagates;

how compensation occurs;

which mechanisms are involved;

and whether a Module's mechanisms are biologically relevant.

The Networks must not become an alternative diagnostic framework or the primary Module selector.

3. Required status of the Network Crosswalk

Every current suggestion should remain:

mapping_status = CANDIDATE

until it is reviewed and approved.

Suggested mappings must not be loaded as official production relationships merely because a code has already been entered.

The Crosswalk should include at least:

network_code

network_number

legacy_display_id

network_name_en

network_name_th

network_scope

primary_axis_code

secondary_axis_codes[]

mapping_reason

mapping_confidence

mapping_status

reviewed_by

review_date

axis_master_version

network_registry_version

4. One Primary Axis and optional Secondary Axes

Each Living Biological Network must have:

exactly one approved Primary Axis;

zero or more Secondary Axes;

an explicit biological reason for the mapping;

a confidence value;

an approval status.

The Primary Axis represents the Network's principal biological home.

Secondary Axes represent meaningful cross-system effects, but they must not displace the Primary Axis.

The spreadsheet should not use several ambiguous Axis columns without clearly stating which one is primary.

Recommended structure:

primary_axis_code

secondary_axis_codes[]

mapping_reason

mapping_confidence

mapping_status

5. Mapping must be semantic, not numerical or lexical

The Network-to-Axis mapping must be based on:

biological function;

mechanism;

Network scope;

upstream and downstream role;

primary biological consequence;

the official definition of the Axis.

It must not be based only on:

the Network number;

a similar word in the Axis name;

the Domain label from the old file;

the organ named in the Network;

a disease association;

or an LLM-generated semantic guess.

A Network may have a name that sounds like one Axis but functionally belongs under another Axis.

6. Confirmed examples from the draft

The following examples shown in the Developer sheet are biologically appropriate:

N033 Phase I Detoxification

Primary Axis: A15 — Detoxification Capacity

N048 Base Excision Repair

Primary Axis: A11 — DNA Repair & Genomic Stability

N053 DNA Methylation

Primary Axis: A12 — Epigenetic Regulation

DNA Methylation must not remain unmapped. It belongs directly under A12.

7. Important candidate mappings that require correction

Several visible suggestions appear to require semantic correction.

7.1 Glucose Variability

If Glucose Variability has been mapped primarily to DNA Repair or another unrelated Axis, it must be corrected.

Recommended mapping:

Primary Axis:

A5 — Glucose–Insulin Regulation

Possible Secondary Axes, depending on the Network scope:

A8 — Mitochondrial Network

A9 — Oxidative Stress & Redox Balance

A27 — Glycation / Carbonyl Stress

Glucose variability is primarily a glucose-regulation problem, even though it may produce oxidative, mitochondrial, and carbonyl consequences.

7.2 DNA Damage Sensing

DNA Damage Sensing should be mapped primarily to:

A11 — DNA Repair & Genomic Stability

It should not be mapped primarily to connective tissue, structural integrity, or another unrelated Axis.

Secondary relationships may exist, but A11 is its principal biological home.

7.3 Fatty-Acid Oxidation and Beta-Oxidation

Fatty-acid oxidation is a mitochondrial energy process.

Recommended mapping:

Primary Axis:

A8 — Mitochondrial Network

Secondary Axis:

A6 — Lipid–Lipotoxicity

When the Network scope focuses specifically on systemic lipid burden rather than mitochondrial oxidation, A6 may be primary. The scope field must determine the final decision.

7.4 Cardiac Networks

Cardiac Networks must not all be placed under A20 merely because vascular function is involved.

They must be separated by mechanism.

Whole-organ cardiac function or reserve

Primary Axis:

A35 — Visceral Organ Functional Reserve

Endothelial or vascular-perfusion mechanism

A20 — Endothelial Function

A21 — Vascular Remodeling & Microcirculation

Hemodynamic or arterial-pressure mechanism

A23 — Hemodynamics & Arterial Elasticity

Cardiac mitochondrial energy

Primary or Secondary Axis:

A8 — Mitochondrial Network

Autonomic or neurocardiac regulation

A34 — Neuro–Sleep–Stress–Brain Clearance

The Axis must reflect the Network's mechanism, not merely the organ name.

7.5 Brain, memory, cognition, and synaptic Networks

Networks involving:

synaptic plasticity;

memory consolidation;

cognitive control;

limbic regulation;

dopaminergic reward;

learning and adaptation;

neuro-autonomic control;

should generally have:

Primary Axis:

A34 — Neuro–Sleep–Stress–Brain Clearance

Possible Secondary Axes may include:

A12 — Epigenetic Regulation

A20 — Endothelial Function

A21 — Vascular Remodeling & Microcirculation

A9 — Oxidative Stress & Redox Balance

A secondary molecular mechanism must not incorrectly replace the neurological Primary Axis.

7.6 Sensory Networks

Networks involving:

vision;

retina;

hearing;

auditory processing;

olfaction;

taste;

local sensory-organ biology;

should generally map primarily to:

A36 — Local Organ & Tissue Interface

A34 may be included as a Secondary Axis when the Network specifically concerns central neural processing rather than the local sensory organ.

7.7 Reproductive Networks

Female and male reproductive Networks should be separated by mechanism.

Female endocrine regulation

Primary Axis:

A31 — Female Hormone Network

Male androgen regulation

Primary Axis:

A32 — Male Androgen Network

Local reproductive-organ or pelvic tissue

Primary Axis:

A36 — Local Organ & Tissue Interface

Vascularization or endothelial support

Secondary Axis:

A20 or A21

Reproductive Networks must not be assigned primarily to A20 solely because vascularization is involved.

7.8 Liver Repair Networks

“Liver Repair” must not automatically be mapped to DNA Repair.

The appropriate mapping depends on scope.

Tissue regeneration or healing

A24 — Wound Healing

Fibrosis and extracellular-matrix remodeling

A25 — Fibrosis & ECM Remodeling

Whole-organ liver reserve

A35 — Visceral Organ Functional Reserve

Detoxification function

A15 — Detoxification Capacity

DNA-repair mechanism inside hepatocytes

A11 — DNA Repair & Genomic Stability

A11 should be used only when the Network scope actually concerns DNA damage or genomic repair.

8. Domain must be derived from the approved Primary Axis

The Network Crosswalk currently contains proposed Domain and Axis columns.

The Domain should not remain an independently editable source of truth after the Primary Axis has been approved.

Use:

`primary_domain_code`

=

`AxisRegistry[primary_axis_code].domain_code`

Examples:

N033 → A15 → Domain 4

N048 → A11 → Domain 3

N053 → A12 → Domain 3

Secondary Domains should also be derived automatically from Secondary Axes.

The `CONFIRM_domain_num` column may remain as a temporary expert-review field, but the production Domain assignment must be generated from the Axis Registry.

This avoids:

duplicate editing;

stale Domain assignments;

conflicting Axis and Domain records;

and accidental inconsistency after an ontology update.

9. Mapping confidence must not become a clinical score

The values shown in the draft, such as:

0

0.17

0.25

0.33

0.50

0.67

1.00

must be explicitly named:

mapping_confidence

They must not be interpreted as:

patient Network activity;

Axis severity;

Module Match;

biological importance;

or clinical confidence.

The development team should formally define the scale.

Example:

1.00 = exact semantic match

0.67 = strong candidate requiring confirmation

0.50 = plausible but not definitive

0.33 = ambiguous

0.17 = weak candidate

0.00 = no safe mapping proposed

These values are workflow tools for expert review only.

They must not affect a patient report or Module ranking until the relationship is approved.

10. Network Code must remain separate from Network Number

The Registry should distinguish:

network_number

network_code

legacy_display_id

Example:

network_number = 33

legacy_display_id = N033

network_code = LBN-DET-PHASE1

network_number is the display order.

network_code is the permanent identity used by:

APIs;

database relations;

Network edges;

reports;

research mappings;

and version history.

The AI must not infer meaning from the code or number. All biological meaning must be retrieved from the approved Registry.

11. The Network Crosswalk does not yet complete the Network Layer

The current Crosswalk appears to address:

Network → Axis

Network → Domain

It does not yet fully define:

Network → Network

The final Network Layer requires a separate Network Edge Registry.

Required fields:

source_network_code

target_network_code

edge_type

direction

biological_effect

context

evidence_status

evidence_grade

source_reference

confidence

mapping_status

version

Recommended Edge types:

ACTIVATES

INHIBITS

REQUIRES

COMPENSATES

PROPAGATES_TO

RESTORES

FAILURE_PRECEDES

ASSOCIATED_WITH

Each Edge must state whether it is:

causal;

mechanistic;

compensatory;

clinically associated;

or hypothetical.

The AI may propose Candidate Edges but may not publish them as Official Edges without expert approval.

12. Review of the Axis Minimum-Evidence sheet

What is correct

The visible A1–A39 list appears consistent with the current official Master.

It correctly includes:

A35 — Visceral Organ Functional Reserve;

A36 — Local Organ & Tissue Interface;

A37 — Oncologic Cell Fate;

A38 — Tumor Microenvironment & Metastatic Dynamics;

A39 — Genomic Regulation Homeostasis.

It does not retain CELA as an Axis.

This part is structurally correct.

What must be corrected

Delete the A0 example row

The row:

A0 — example — delete

must be physically removed before the file is loaded.

It must not merely be hidden, because a loader may still ingest it as a real Axis.

The sheet is still a template

The Minimum-Evidence fields have not yet been completed.

Current status should therefore be:

status = TEMPLATE

authoritative = false

production_allowed = false

Until the rules are completed, the Engine cannot determine whether an Axis is assessable.

13. Low, Medium, and High are not enough for Minimum Evidence

A single minimum_tier field may be retained for human readability, but the Engine requires structured rules.

Recommended schema:

axis_code

core_evidence[]

supporting_evidence[]

optional_evidence[]

minimum_core_count

minimum_total_weight

required_modalities[]

accepted_inputs[]

exclusion_rules[]

allow_provisional_assessment

status

version

Previously agreed evidence weights should remain:

CORE = 3

SUPPORTING = 2

OPTIONAL = 1

Minimum Evidence must be specific to each Axis.

For example:

A26 may require at least a CBC or equivalent hematological evidence.

A8 may accept a structured energy phenotype as candidate evidence, but high-confidence mitochondrial assessment requires objective support.

A37 requires verified oncology evidence such as pathology, imaging, or clinician-confirmed cancer information.

A19 requires colon-specific evidence and must not be inferred from constipation alone.

14. Required Axis assessment states

The Engine should use:

NOT_ASSESSED

AXIS_CANDIDATE

PROVISIONALLY_ASSESSED

ASSESSED

HIGH_CONFIDENCE_ASSESSED

The decision rule is:

Insufficient evidence

→ NOT_ASSESSED

not:

Insufficient evidence

→ 0

→ 50

→ Mild

Missing data must never be treated as normal.

15. Required production behavior

The Network and Minimum-Evidence files must not influence clinical recommendations until they pass approval gates.

Recommended production rules:

Network mapping not approved

→ Network remains NOT_ASSESSED

Axis minimum evidence not defined

→ Axis cannot receive an authoritative score

Module mapping not approved

→ Module cannot enter clinical ranking

Domain conflict

→ derive Domain from approved Primary Axis

Mapping confidence only

→ do not use in patient scoring

16. Recommended workflow for correcting the files

Step 1 — Clean the Axis template

delete A0;

freeze A1–A39 names and Domains;

add the Axis Master version;

lock protected columns;

complete structured Minimum-Evidence fields.

Step 2 — Review high-confidence Network mappings

Begin with mappings that are biologically direct, such as:

Phase I Detoxification → A15;

Base Excision Repair → A11;

DNA Methylation → A12.

Step 3 — Review ambiguous mappings

Prioritize rows with:

mapping_confidence < 0.67

and rows involving:

cardiac systems;

neurocognitive Networks;

liver repair;

reproductive systems;

sensory systems;

organ reserve;

oncology;

cross-domain mechanisms.

Step 4 — Separate Primary and Secondary Axes

Do not keep ambiguous “Axis 1 / Axis 2” columns.

Use permanent Primary and Secondary fields.

Step 5 — Derive Domains

After the Primary Axis is confirmed, calculate the Domain from the Axis Registry.

Step 6 — Create the Network Edge Registry

Add approved upstream/downstream biological relationships with Edge types and evidence.

Step 7 — Expert approval

Every approved row should record:

reviewed_by

review_date

mapping_status

axis_master_version

network_registry_version

Step 8 — Validation testing

The Loader must reject:

unknown Axis codes;

unknown Network codes;

multiple Primary Axes;

invalid Domain–Axis combinations;

missing Registry versions;

unapproved mappings in production;

duplicate permanent identifiers.

17. Final status assessment

Component

Current assessment

Current A1–A39 list

Structurally correct

Domain list

Structurally correct

A0 example row

Must be deleted

Axis Minimum-Evidence content

Incomplete

180 Network extraction

Useful and necessary

Suggested Network mappings

Candidate only

Primary/Secondary Axis design

Needs clarification

Domain mapping

Should be derived from Axis

Mapping confidence

Review aid only

Network Edge Registry

Still required

Ready for production

No

Final instruction to the development team

The spreadsheets are well-designed as review instruments, but they must not yet be treated as approved TSPI knowledge registries.

Please retain all Network mappings as CANDIDATE until semantic review is complete.

Use the biological mechanism and official Axis definition—not the Network name, number, old Domain label, or lexical similarity—to determine the Primary Axis.

Each Network must have exactly one approved Primary Axis, optional Secondary Axes, a mapping reason, mapping confidence, version, and approval record.

Derive Domains automatically from the approved Axis Registry.

Do not use mapping-confidence values as clinical scores or Module-Match values.

Complete the Network Edge Registry before considering the 180-Network layer operational.

Delete the A0 example row and complete the Minimum-Evidence rules before the Axis sheet is loaded into production.

When evidence is insufficient, the correct output is NOT_ASSESSED. Never replace missing evidence with zero, normal, Mild, or a default value of 50.

The Thirty-Nine Biological Axes remain the centre of TSPI. The Networks explain the Axes; they do not replace them.